GROCERY STORE MANAGEMENT SYSTEM (SDD)

SOFTWARE DESIGN DOCUMENT

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The Islamia University of Bahawalpur

DEPARTMENT OF COMPUTER SCIENCE

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Revision History

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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Sir Nasir Mirza	Supervisor, CSIT 21306	<date>

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CHAPTER 1: INTRODUCTION

Store is a place where access material is kept which will be used as and when required. Loss of items, deterioration, obsolescence and inadequacy are treated “as a part of life “store management system ”is to receive material” to protect them while in storage from damage& unauthorized removal to issue the material in the right quantities, at the right time to the right places and provide to these service promptly and at least cost.

1.1 Purpose

The purpose of the document is to collect and analyze all the ideas. So we want to sort out this product will be used in to order in gain a better understanding of the project develops. In short, the purpose of this SRS document is providing a detailed overview of our software product its parameter and goals. This document describes the project’s audience and its user interface, hardware, software requirements. It defines its functionality. None the less, it helps to design software delivery life cycle processes.

1.2 Scope

In this subsection:

- Minimize cost of production through minimize cost on materials.
- Maintain the value of materials.’
- Service to user departments.
- Establishing co-ordination with other departments.
- Advising materials manager.

1.3 Overview

This document provides a general description, including features of the projects. The products of hardware, data requirements and functional of the products. General description of the project is discussed in section 2 of this document. Section 3 will explain the detailed functional specification system and other requirements of the system.

1.4 Reference Material

In this subsection

(1) www.google.com.
(2) www.draz.pk.
(3) www.wikipiadia.com
(4) www.3school.com.

1.5 Definitions and Acronyms

- WWW (worldwide web set of large interlinked document on internet).
- SQL (Structure Query Language).
- GUI (Graphical User Interface).
- Modify (for modification in database or user record).
- Admin (Login for attraction with application).
- Logout (Logout for UN attraction with application).
- RESUME (To restarting something).
- PAUSE (To stop something).
- Visual Studio: is an IDE that support multiple languages like c#, java.net.
- Frame Work: is a platform for developing application. It is launched by Microsoft. It gives features for developing application.
- DFD: is a stand for data flow diagram.
- ER Diagram: stands for entity relationship diagram.

CHAPTER 2: SYSTEM OVERVIEW

2.1 Technologies Used

The system is coded with c# by using VISUAL STUDIO Tool. We will use service base database.

2.2 Application Overview

The main objective of the project is sale items.

This “” consist on two parson is admin site and second one is user site .user can visit and buy product by using personal account. Whereas admin can add, delete product update stock and price etc by authentication. Both parsons can check delivery status of order.

- Admin must login for any modification
- If customer wants to buy anything more then he just login and fulfill our order form and click order button.
- All required record of customer will save in database.
- Customer can sending by Gmail or contact us with our given cell no etc.
- When order is submit you will see message.

2.3 Design Languages

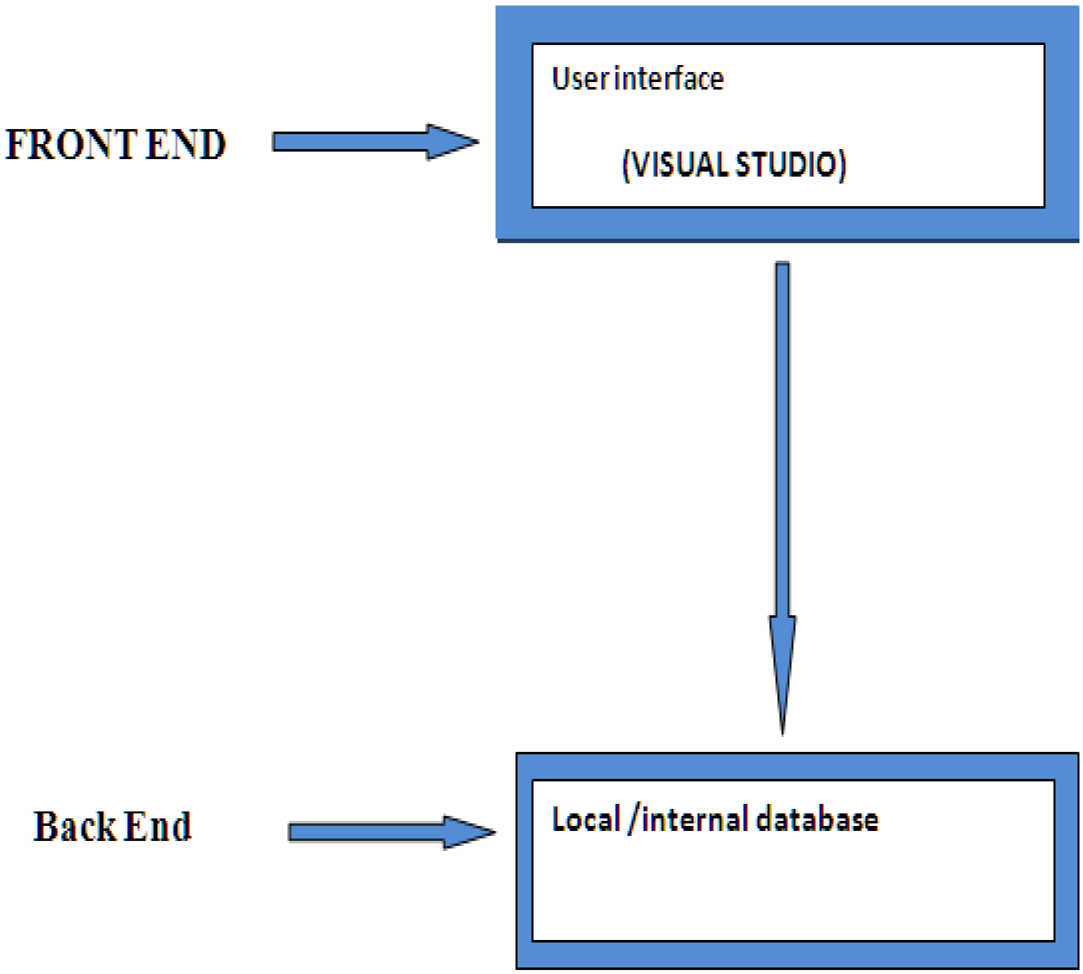
- C-sharp(c#)
- Adobe Photoshop

CHAPTER 3: SYSTEM ARCHITECTRE

3.1 Architectural Design

The interface of the system will design in c-sharp language by using tool VISUAL STUDIO.

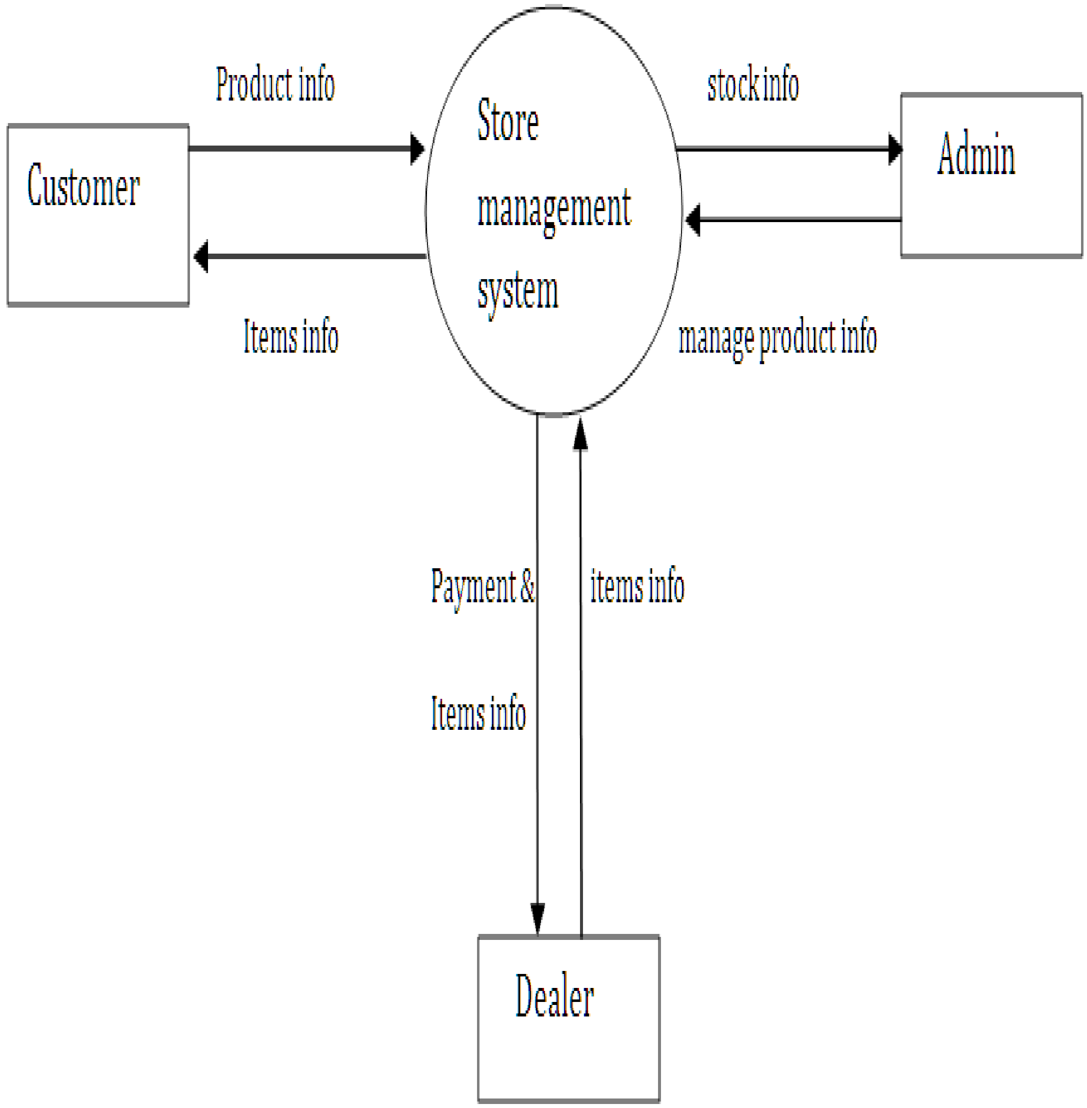
This application will run on a single personnel PC in a shop because it developed as a desktop application. All of the record and entries are stored in local/internal /offline database.



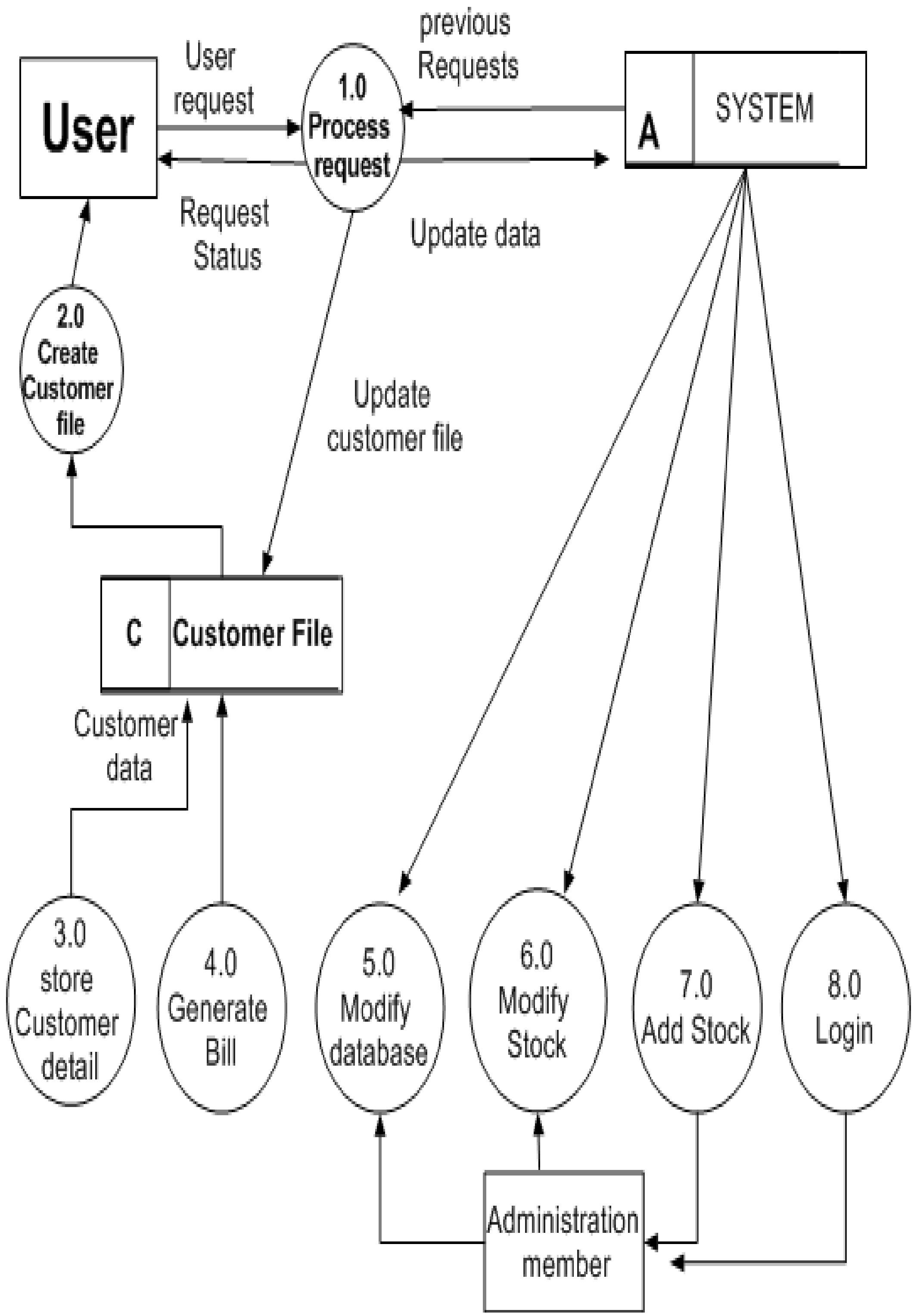
3.2 Decomposition Description

Context level:

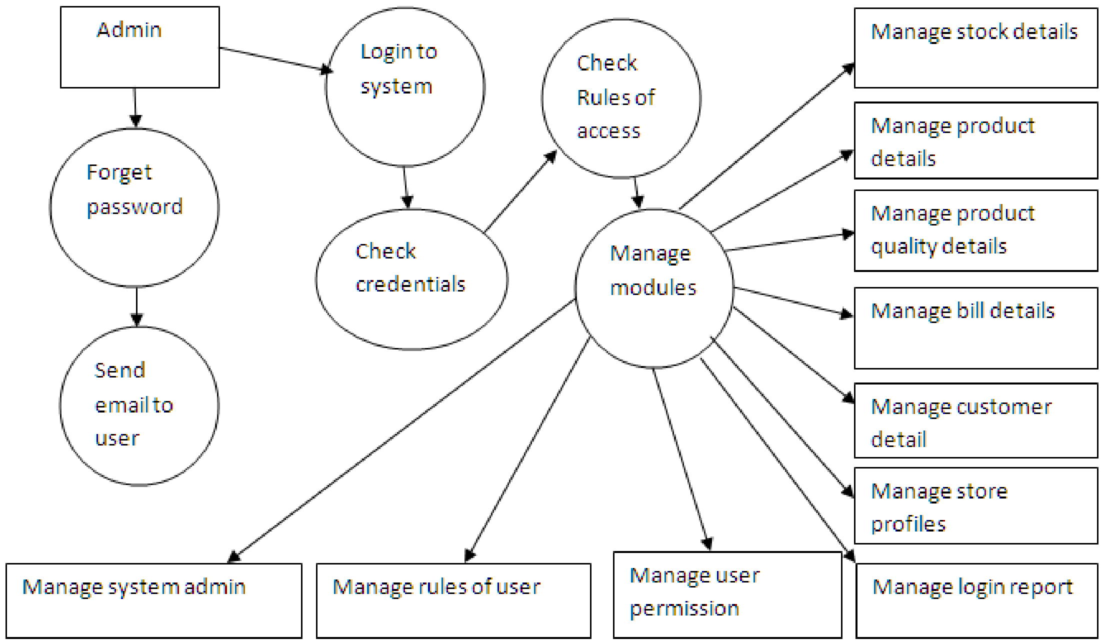
Context level is the top level dataflow diagram. It only contains one process node that generalizes the function of the entry system in relationship to external entities. A context level diagram is drawn in order to define and clarify the boundaries of the software system .It identifies the flows of information between the system and external entities.



DFD (0-level):



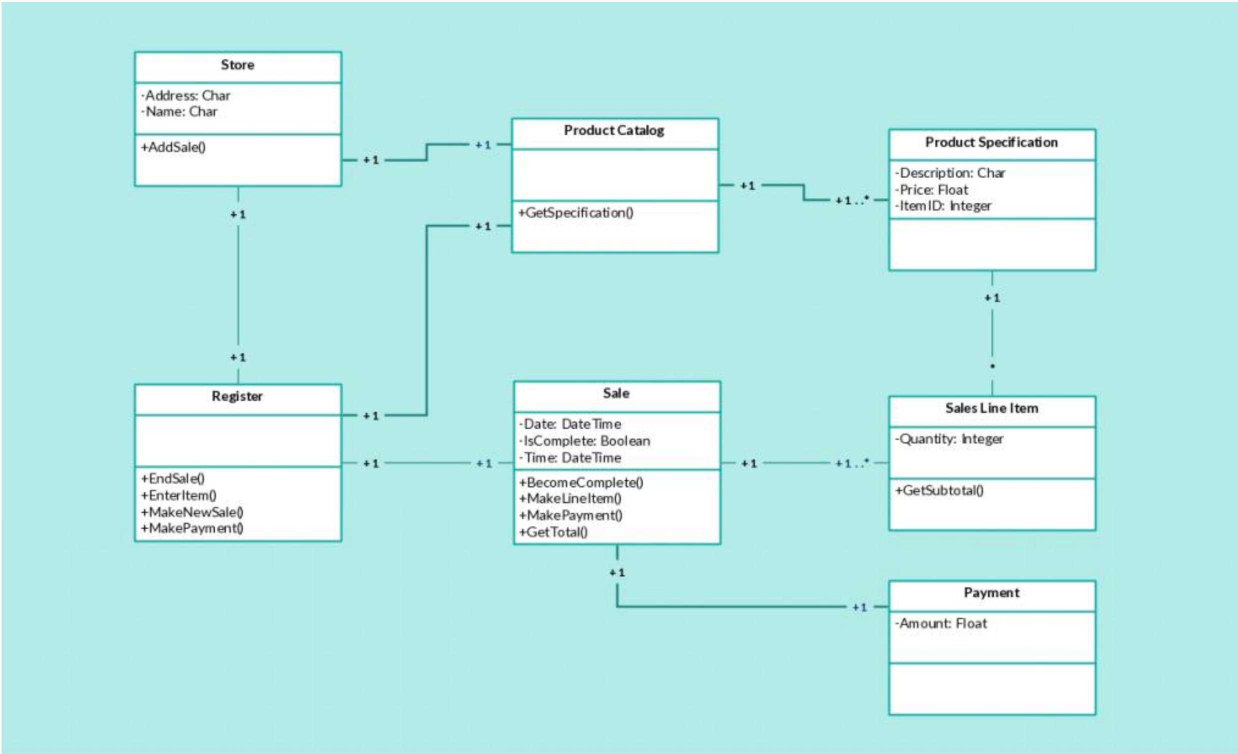
Detail level:



Class Program

The static diagram which represents the static view of an application. It is known as "**class Program**"

A class's attributes operations and the systems are constraints are describing by the class. Due to their ability to be mapped directly with object oriented language, it is used for modeling sub systems. Also known as a structural diagram, it is collection of constraints, associations. In a class diagram, it is necessary that there exists relationship between the classes. The similarity of various relationships often makes it difficult to understand it. Below are the relationships which exist in a class diagram.



Activity Diagram:

Activity diagram is graphical representation of workflows of stepwise activity and action with support for choice, iteration and concurrency. Activity diagram are often used in business process modeling. They can also describe the steps in a use case diagram. Activities modeled can be sequential and concurrent. In both cases an activity diagram will have a beginning (initial state) and an end (a final state).

Activity diagram are used for

1) Documenting exciting process
2) Analysis new process concept
3) Finding reengineering opportunities.

Basic Activity Diagram Notations and Symbols:

Initial state and ending point:

A small field circle follow by an arrow represent the initial action state or the start point for any activity diagram.

Start point /initial state

Activity:

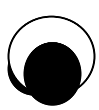
An action state represents the non-interrupted action of objects. You can draw an action state in smart draw using a rectangle with round corners.

Action flow:

Action flows, also called edge and path, illustration and transition from one action state to another. They are usually drawn with an arrowed line.

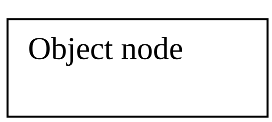
Activity final Node:

Stop all control flows and object flows in an activity.



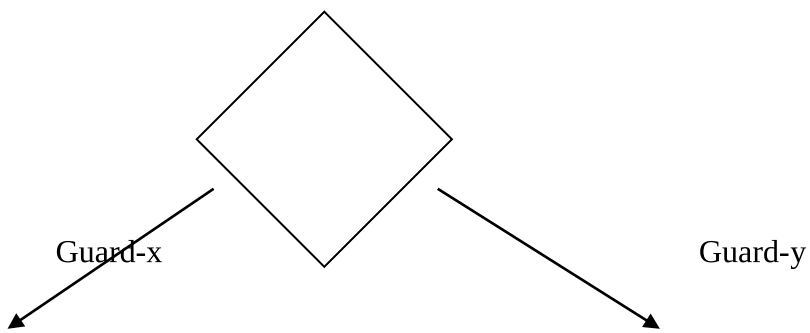
Object node:

Represent an object that is connected to a set of a object flow.



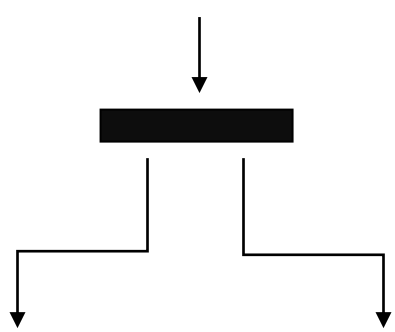
Decision Node:

Represent a test condition to ensure that the control flow or object flow only goes down one path.



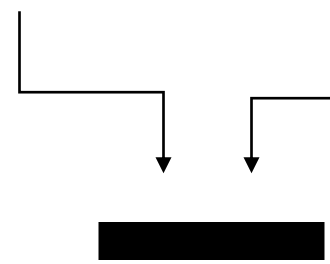
Fork Node:

Fork node is a control node that has one incoming edge and multiple outgoing edges and is used to spilt incoming flow into multiple concurrent flows.

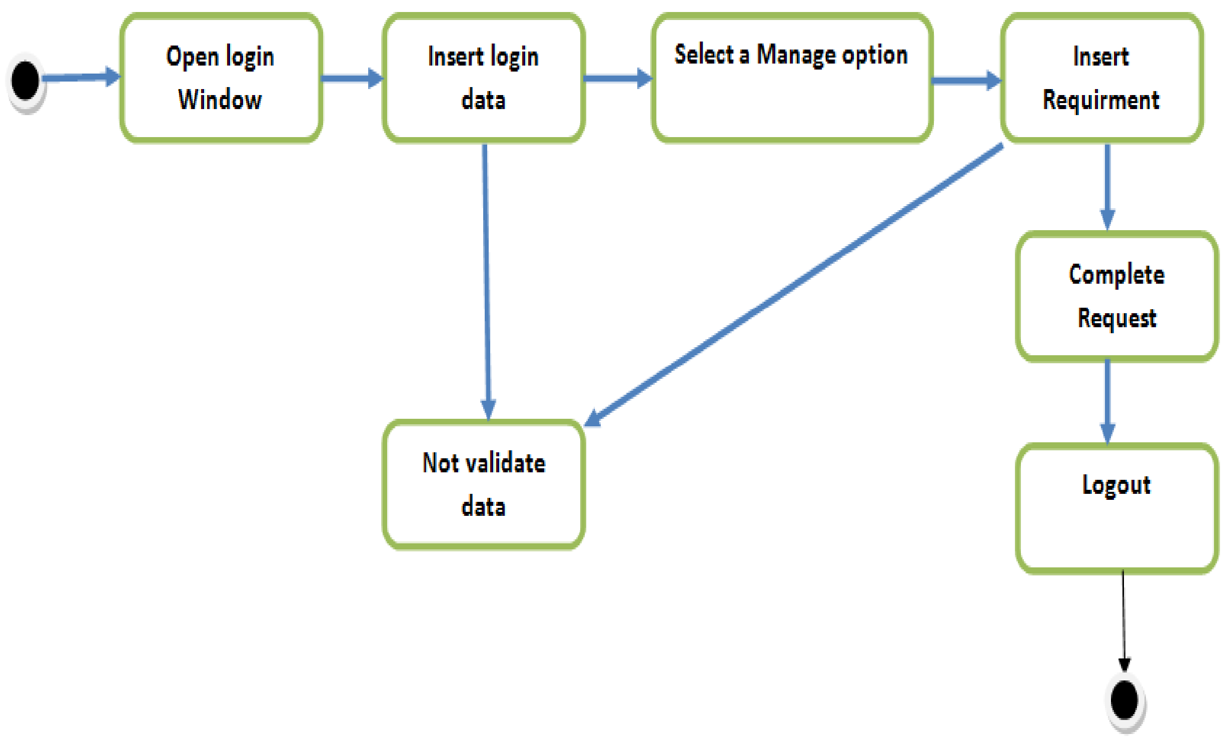


Join Node:

Bring back together a set of parallel or concurrent flows of activities.



Activity Diagram:



3.3 Design Rationale

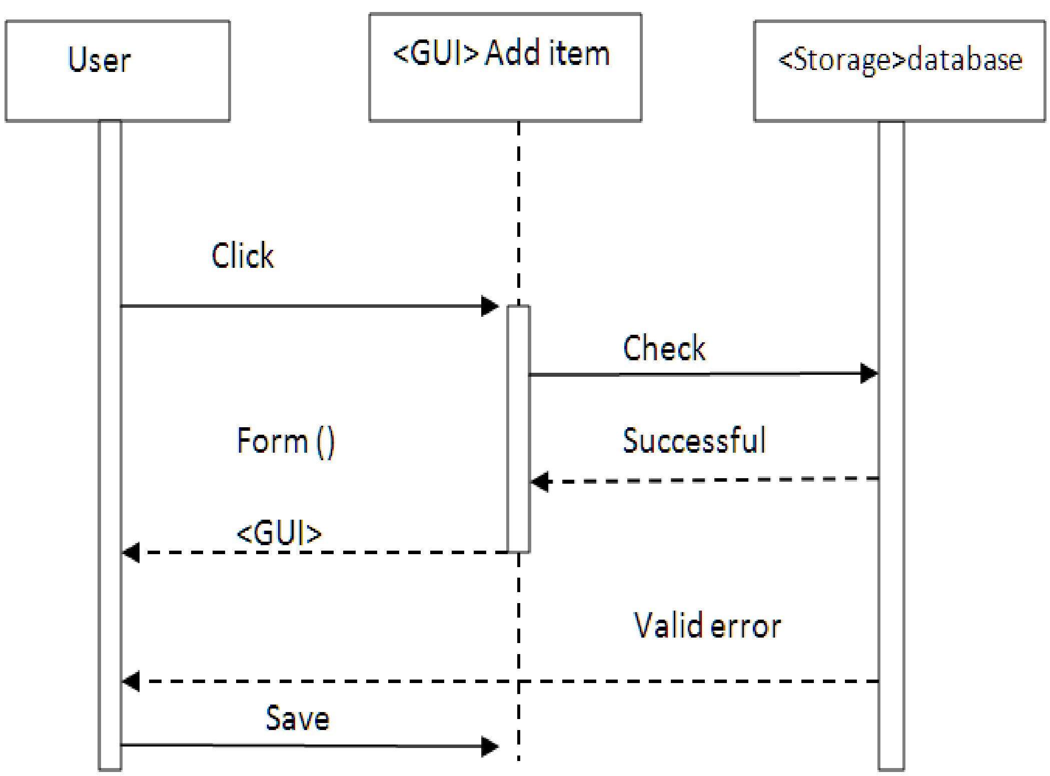
This application **“Store management System**” is a desktop application which will be developed /coded in C-Sharp1(c#) language and the tool that is used for this language is **VISUAL STUDIO.** This tool is easy for developed this application. To records or save data we use the internal/local/offline database by connecting. So, the storing activity performed by the local database.

- **Sequence Diagrams:**

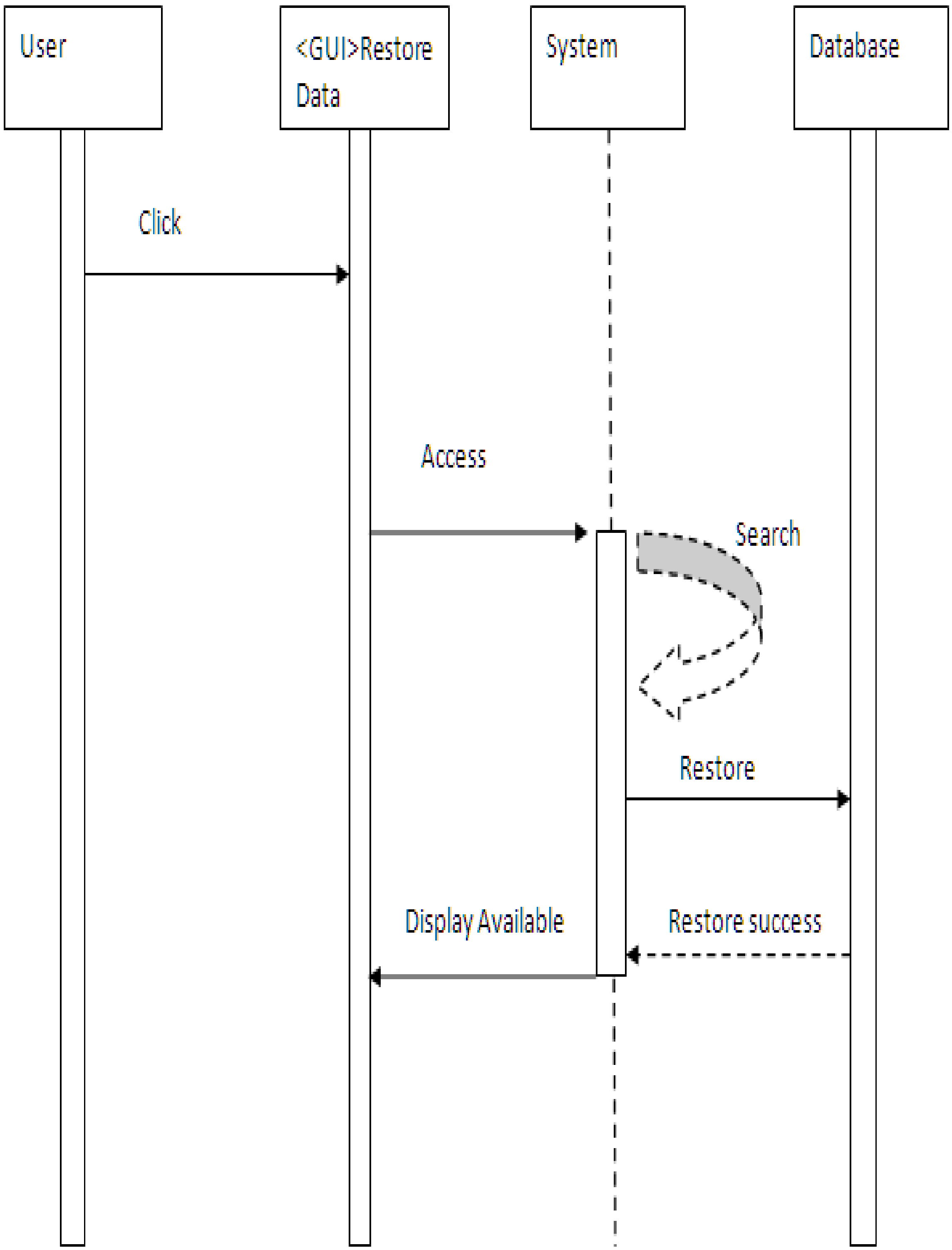
Sequence diagram are interaction diagrams that detail how operations are carried out

- Interaction diagrams model important runtime interaction between the parts that makes up system.
- A sequence diagram is made up of a collection of participants.
- Class or object- each class in the interaction is represented by its name icon along the top of the diagram.

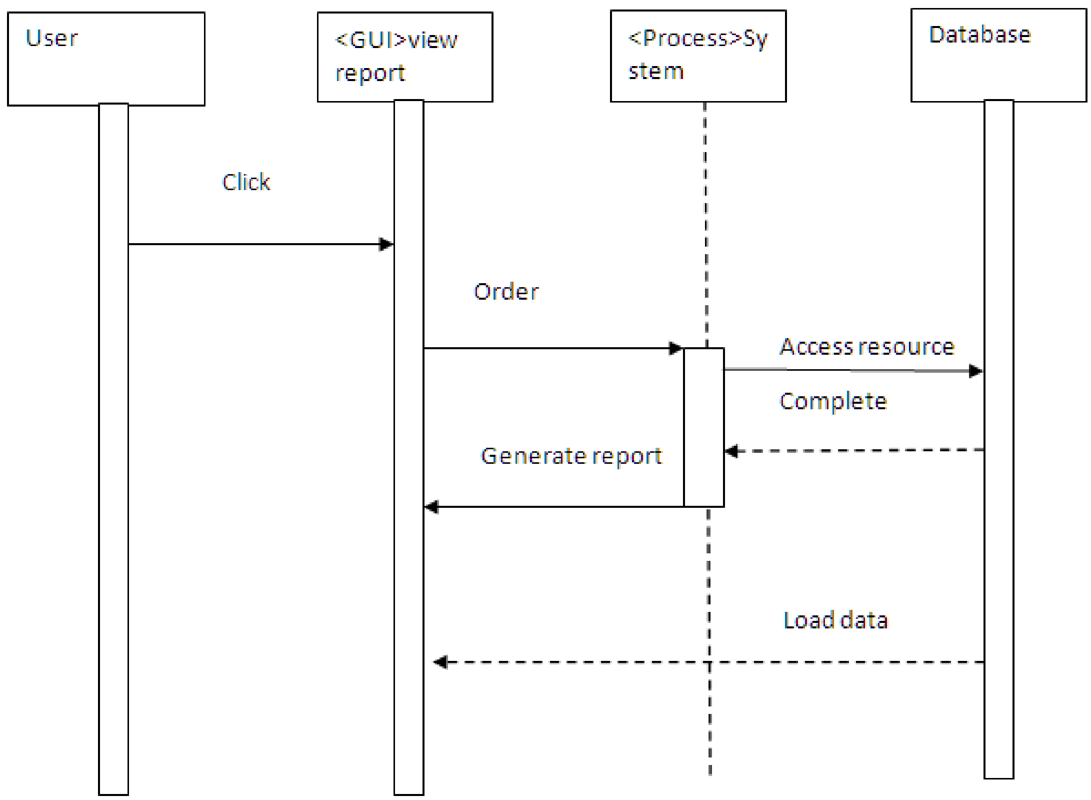
Add Item:



Sequence diagram for restore data:



Sequence diagram for view Report:

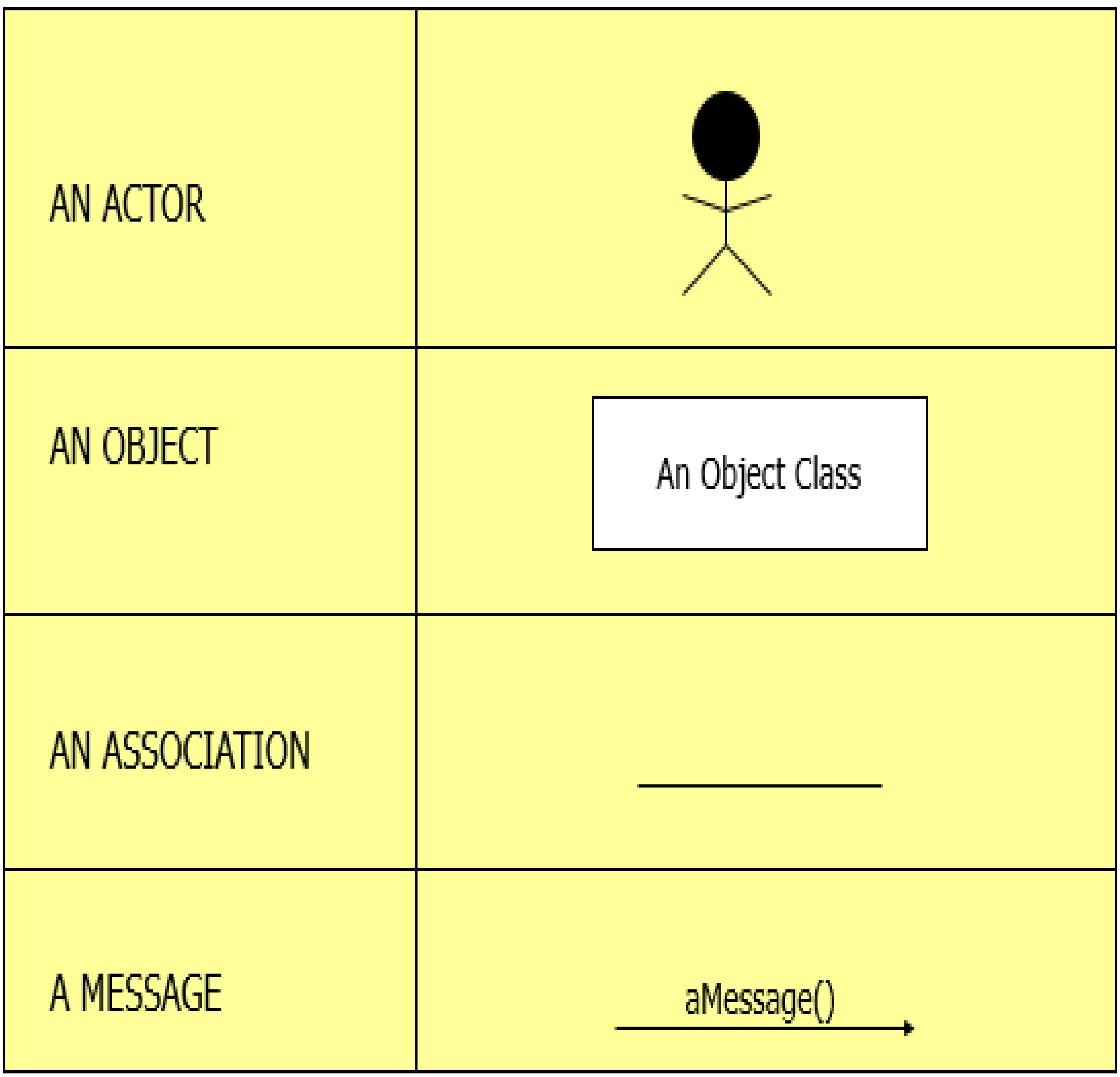


Collaboration Diagram

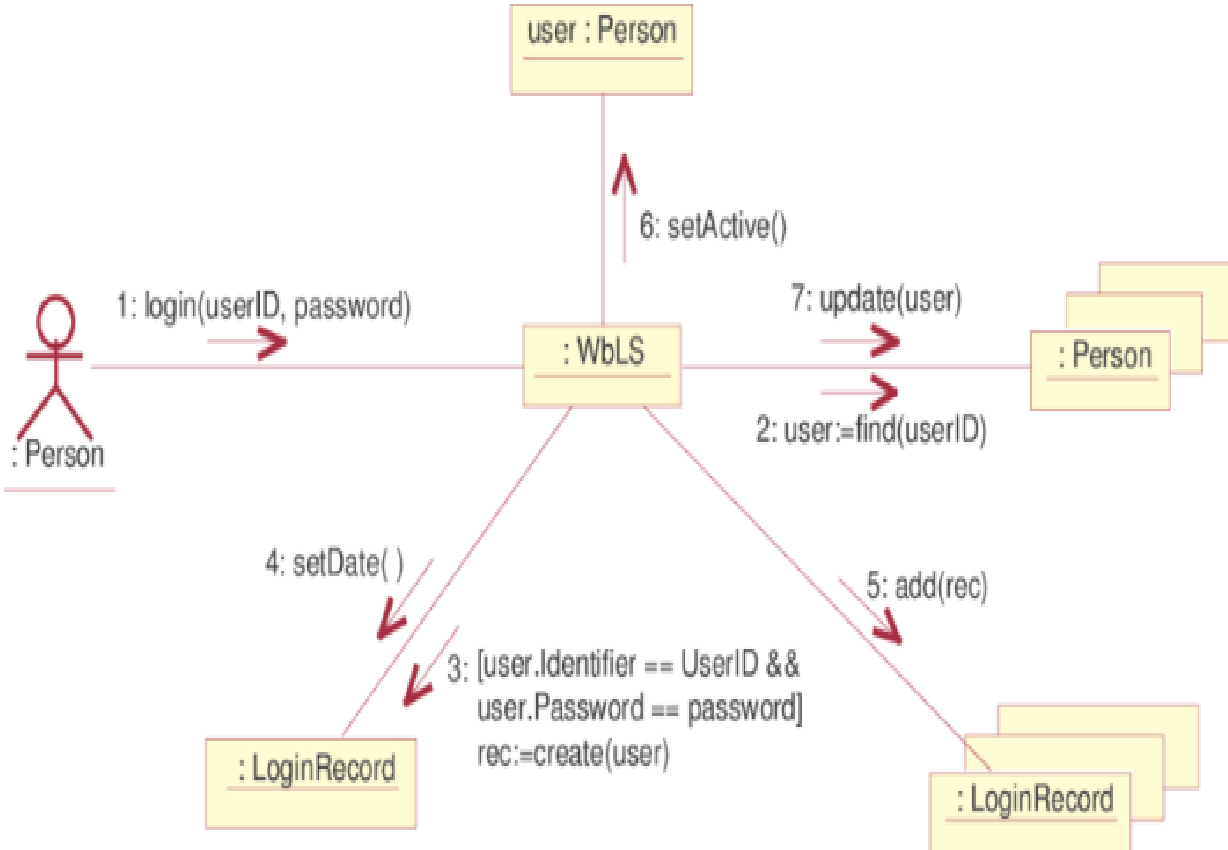
- Collaboration diagrams illustrate interactions between objects
- The collaboration diagram illustrates messages being sent between classes and objects (instances).
- Collaboration diagrams express both the context of a group of objects (through objects and links) and the interaction between these objects (by representing message broadcasts)
- They are very useful for visualizing the relationship between objects collaborating to perform a particular task
- It provide a good interaction between objects which may be difficult to see at the class level
  There are three primary elements of a collaboration diagram:
- Objects
- Links
- Messages

So we can create a collaboration diagram by:

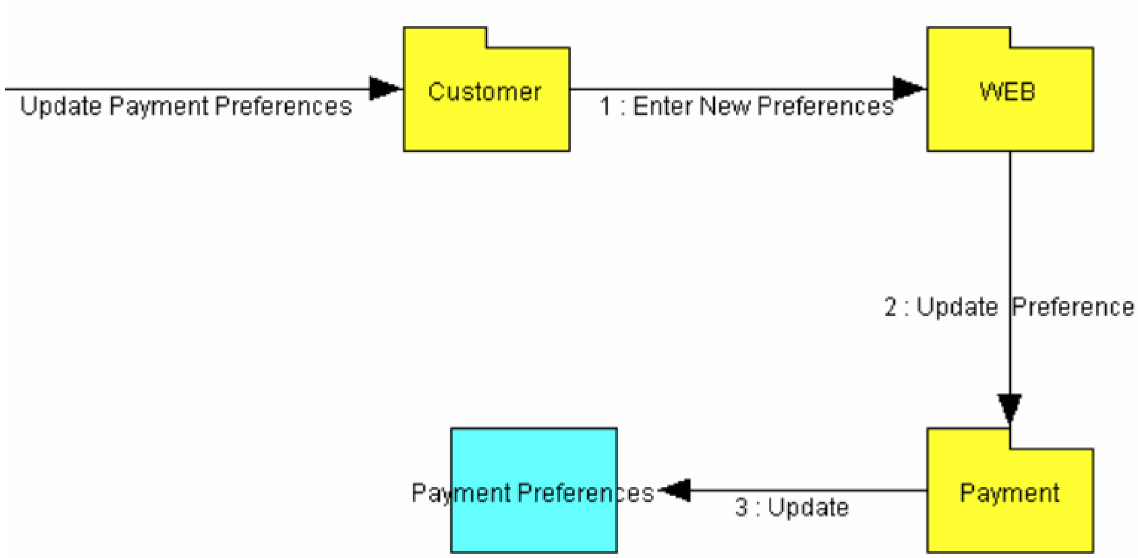
- Create links between the objects
- Create messages associated with each link
  - Add sequence numbers to each message corresponding to the time-ordering of messages in the collaboration
  - **Collaboration Diagram Syntax**



- **Collaboration Login**



Collaboration payment and update



CHAPTER 4: DATA DESIGN

4.1 Data Description

1. Different constraints/checks are applied
2. Required data of customers as condition stored in the database. All of this data will be accurate which stored in database.
3. We choose C-Sharp**(C#)** language for developing or designing this application.
4. **VISUAL STUDIO** tool is use for this language.
5. **Store management System** is a desktop application and runs on a single PC at shop.

4.2 Data Dictionary

Descriptions of tables are as follows:

Data Dictionary of Store Login table:

- Email
- Admin_ Email
- Administration Email
- Varchar(50)
- 256 Characters
- 0-9 and a-z and A-Z and special symbol

- Password
- Admin_ password
- Administration Password
- Varchar(50)
- 256 Characters
- 0-9 and a-z and A-Z and special symbol

Table:

	Name	Data Type	Allow Nulls	Default
	Username	nvarchar(50)	☐	
	Password	int	☐	
🔑	email	nvarchar(50)	☐	
			☐	

Query of this table:

```
CREATE TABLE [dbo].[AdminLogIn] (
    [Username] NVARCHAR (50) NOT NULL,
    [Password] INT           NOT NULL,
    [email]    NVARCHAR (50) NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY CLUSTERED ([email] ASC)
);
```

Database of logintable:

	Username	Password	email
▶*	NULL	NULL	NULL

Data Dictionary of store record table:

- Model
- model
- item Model
- Varchar
- 256 Characters
- a-z and A-Z

- Collections
- collection
- items stock
- Integer
- 18 digit

- 0-9

- Photo
- photo
- items Picture
- Varchar
- 256 Characters
- 0-9 and a-z and A-Z and special symbol

Table:

	Name	Data Type	Allow Nulls	Default
🔑	Customer Name	nvarchar(50)	☐	
	Customer Id	int	☐	
	Item	int	☐	
	price	int	☐	
	Category	nvarchar(50)	☐	
			☐	

Query of this table:

```
CREATE TABLE [dbo].[Table]
(
    [Customer Name] NVARCHAR(50) NOT NULL PRIMARY KEY,
    [Customer Id] INT NOT NULL,
    [Item] INT NOT NULL,
    [price] INT NOT NULL,
    [Category] NVARCHAR(50) NOT NULL
)
```

Database of store item record table:

	Customer Name	Customer Id	Item	price	Category
*	*NULL*	*NULL*	*NULL*	*NULL*	*NULL*

Data Dictionary of Stock update table:

- Model
- model
- Items Model
- Varchar
- 256 Characters

- a-z and A-Z

- Collections
- collection
- Items stock
- Integer
- 18 digit
- 0-9
- Photo
- photo
- Items Picture
- Varchar
- 256 Characters
- 0-9 and a-z and A-Z and special symbol

Table:

	Name	Data Type	Allow Nulls	Default
	Item Name	nvarchar(50)	☐	
	Item Number	int	☐	
🔑	Item ID	int	☐	
	Item Price	int	☐	
	Item Quantity	int	☐	
			☐	

Query of this table:

```
CREATE TABLE [dbo].[Stock Update]
(
    [Item Name] NVARCHAR(50) NOT NULL ,
    [Item Number] INT NOT NULL,
    [Item ID] INT NOT NULL,
    [Item Price] INT NOT NULL,
    [Item Quantity] INT NOT NULL,
    CONSTRAINT [PK_Table] PRIMARY KEY ([Item ID])
)
```

Database of stock update table:

	Item Name	Item Number	Item ID	Item Price	Item Quantity
*	*NULL*	*NULL*	*NULL*	*NULL*	*NULL*

Data Dictionary of Bill table:

- Date
- Date
- Shopping date
- Varchar
- 256 Characters
- 0-9999999999999999999 and special symbol

- Customer Name
- name
- Name Of customer
- Varchar
- 256 Character
- a-z, A-Z

- Total Bill
- Bill
- Total bill of customer
- Integers
- 18 digits
- 0-9

Table:

	Name	Data Type	Allow Nulls	Default
🔑	Date	date	☐	
	Customer Name	nvarchar(MAX)	☐	
	Total Bill	int	☐	
			☐	

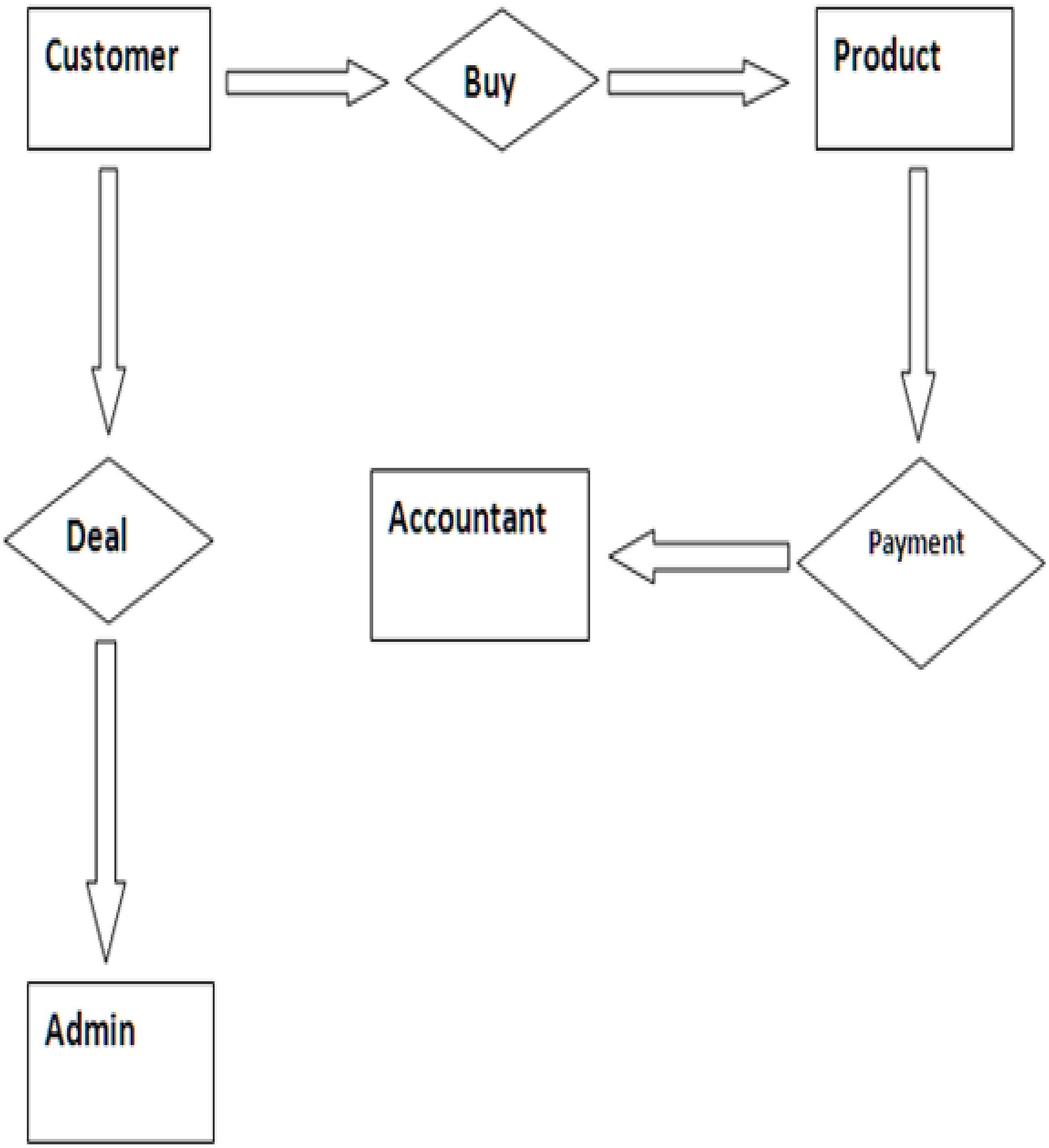
Query of this table:

```
CREATE TABLE [dbo].[Table]
(
    [Date] DATE NOT NULL PRIMARY KEY,
    [Customer Name] NVARCHAR(MAX) NOT NULL,
    [Total Bill] INT NOT NULL
)
```

Database of bill table:

	Date	Customer Name	Total Bill
*	*NULL*	*NULL*	*NULL*

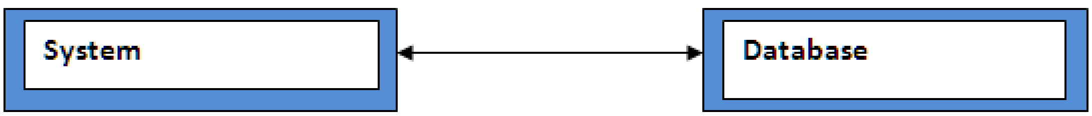
E.R Diagram:



CHAPTER 5: COMPONET DESIGN

5.1 Component design

The system is capable to do work according to the direction. Every simplest or complex Component is responsible to do work which is assigned to it. The system considers every Component as best efficient way component for the mutual cooperation of the whole Processes. Component design follows as:-



Front End **Back End**

The short detail of the component design working describes as the admin after login to the system will choose an option for the submission or other action. Consider an example if admin want to generate bill then admin will press the button of total bill. The given data in the text box will be stored in the database; similarly admin can do updating by the corresponding button.

CHAPTER 6: HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

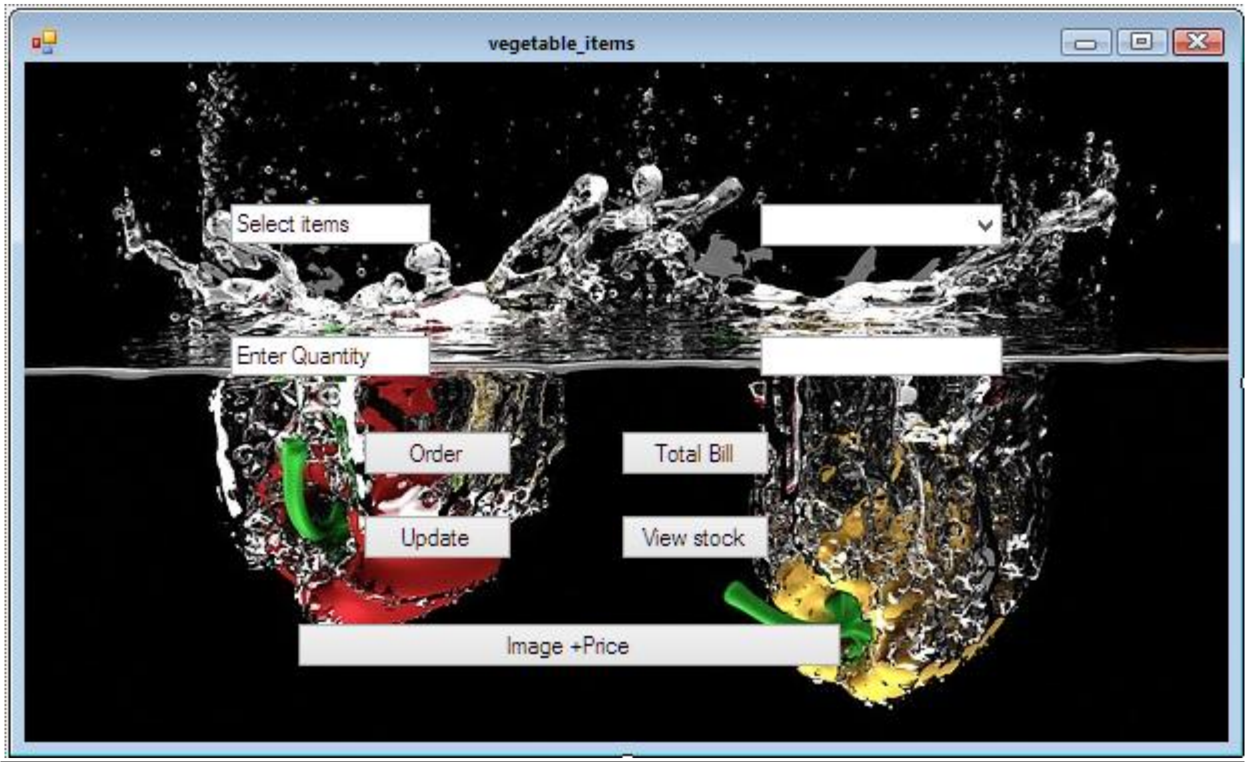
This section describe about the functionality of the different forms

6.2 Screen Images

Login Form

Item Selection Form

Vegetable with Price

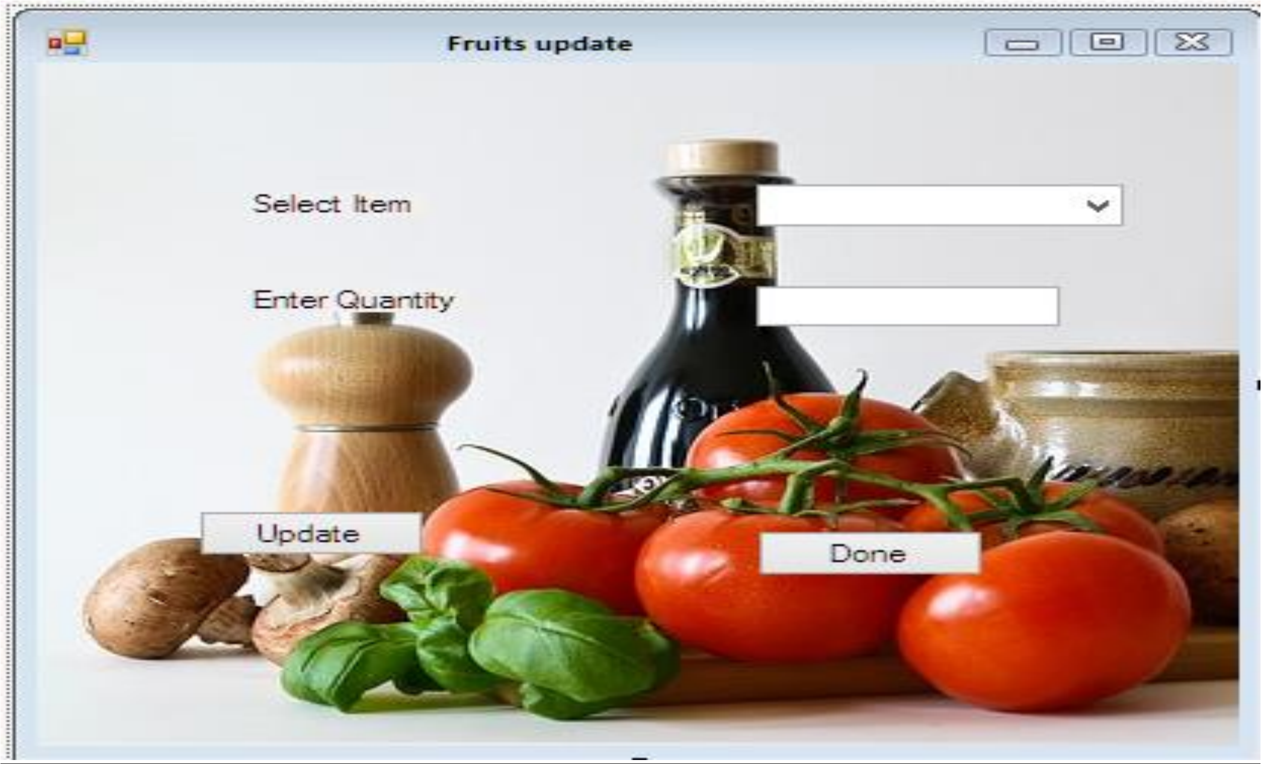


Main form Where All item are shown

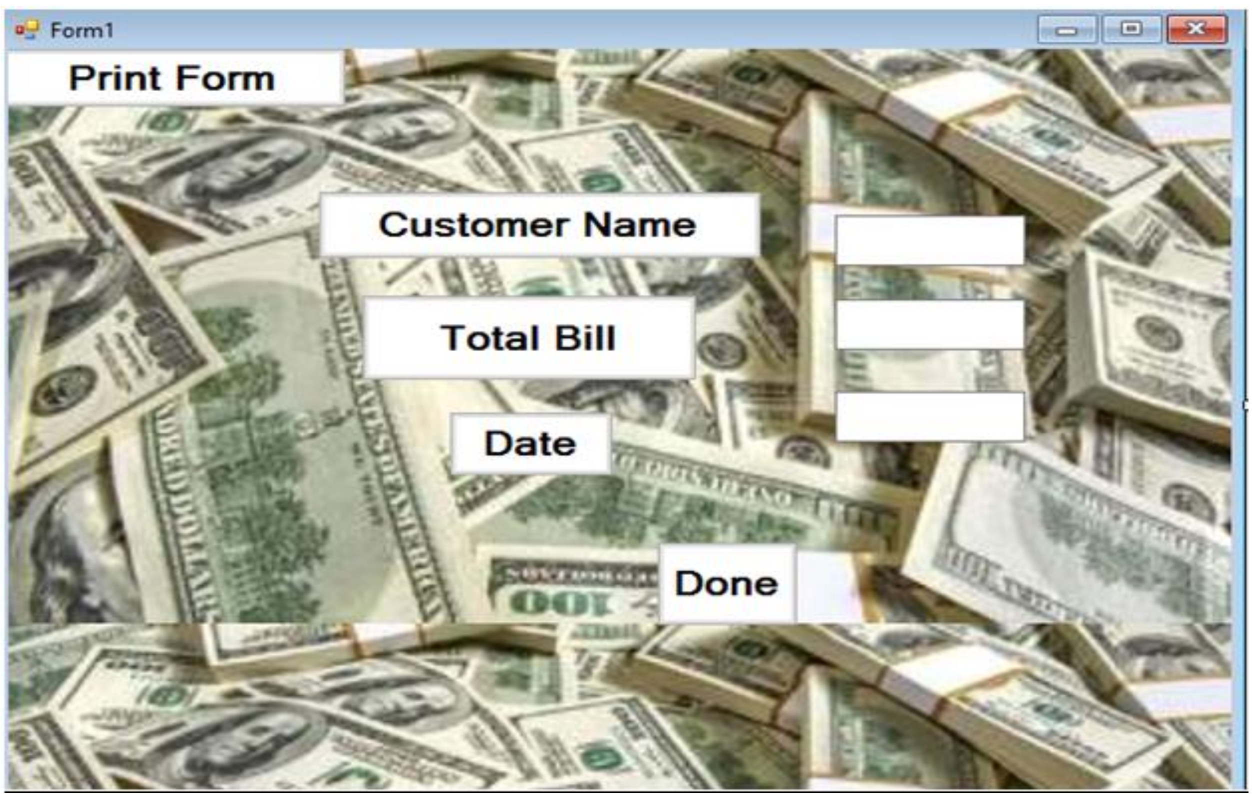
Vegetable update Form

Fruits Main Form

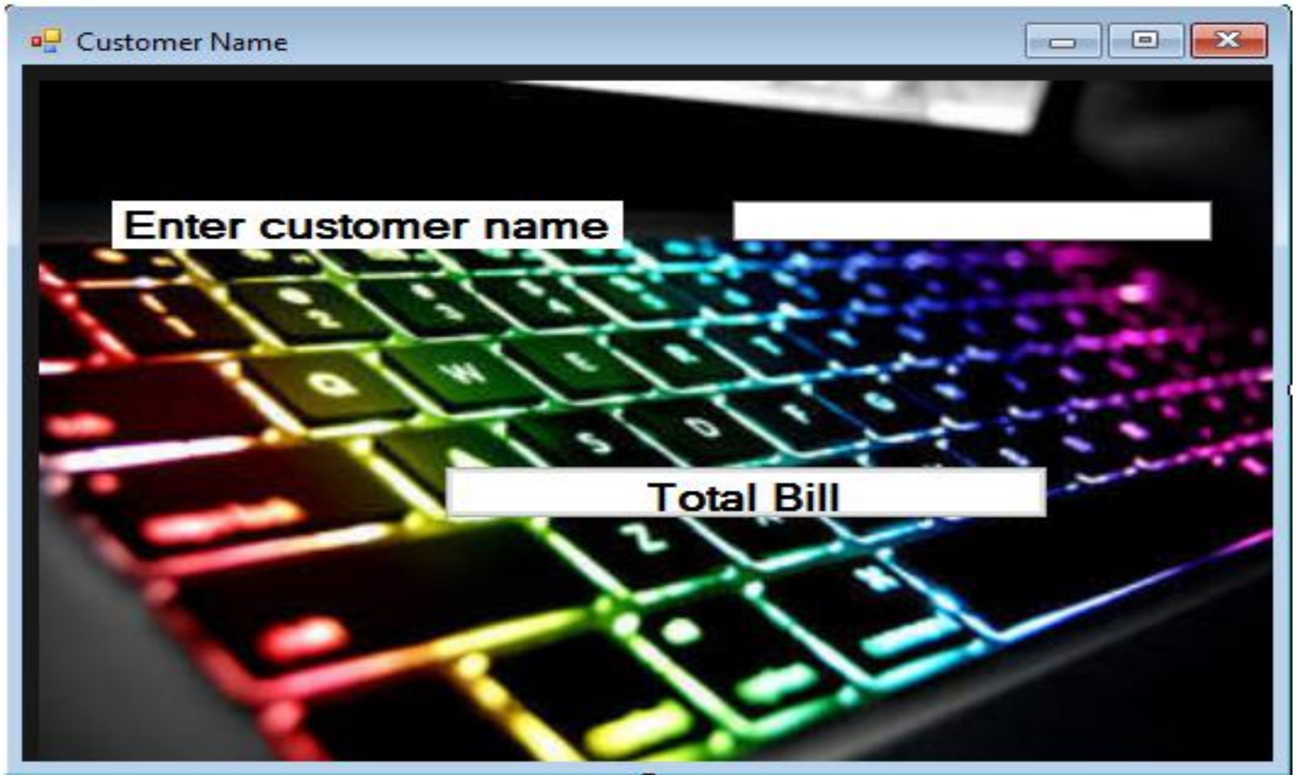
Fruits item with price



Fruit update form



Bill Generation Form



Customer Information Form

Dry fruit Main Form

Dry fruit items with price

Dry fruit update form

6.3 Screen Objects and Actions

A discussion of screen objects and actions associated with those objects.

Login Form:

Login form is used for admin. It will be used for every time when system will shut down and restart this application. This form is necessary for interact with system.

Main/ selection form:

After admin login in this application then this form will be open .This form has a different store items. The admin select the company that customer want to purchase a mobile phones. May be another companies will be mentioned in this form.

Customer info form:

CHAPTER 7: REQUIREMENTS MATRIX

Components: Requirements from SRS (Use case):	Login logout Component	Main Functionality Component	Settings Component	Social Component	Trade Component	Data Services Component	Remote Server Component
UC1	x	x					
UC2	x	x					
UC3	x	x					
UC4	x	x	X				
UC5	x		X				x
UC6		x					
UC7	x	x					
UC8							